

EXAIM Test Data: MCA Big Data Analytics
Subject: MCA305T - Big Data Analytics
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SECTION A (Answer any Four)

1. Explain SVM Concepts in Data Classification

Support Vectors: These are the data points that lie closest to the decision surface (hyperplane). They are the most difficult points to classify and directly define the location and orientation of the hyperplane.

Hyperplane: In an n-dimensional space, a hyperplane is a flat affine subspace of dimension $n-1$ that acts as a decision boundary separating different classes of data.

Kernel: A mathematical function used to transform input data into a higher-dimensional space where it becomes easier to segregate classes that are not linearly separable in the original space.

Hard Margin: A strict approach where the SVM seeks to find the maximum margin hyperplane that perfectly separates all data points without any misclassifications.

Soft Margin: An approach that allows for some misclassifications or points within the margin to handle noisy data and achieve better generalization.

2. MapReduce: Matrix-Vector Multiplication

Map Phase: For each element M_{ij} of the matrix, the mapper produces a key-value pair $(i, M_{ij} \times v_j)$.

Reduce Phase: For each key i , the reducer sums all the values associated with that key to produce the i -th element of the resulting vector.

3. Regression Modeling Methods

Linear Regression: Models the relationship between a dependent variable and one or more independent variables using a straight line.

Logistic Regression: Used for binary classification problems where the output is a probability between 0 and 1.

Polynomial Regression: Models non-linear relationships by fitting a polynomial equation to the data points.

5. Kernel Methods in SVM

Linear Kernel: The simplest kernel; used when the data is already linearly separable. Formula:

$$K(x,y)=x \cdot y.$$

Polynomial Kernel: Represents similarity of vectors in a feature space over polynomials.

Gaussian (RBF) Kernel: Maps samples into infinite-dimensional space, handling complex class boundaries.

SECTION B (Answer any One)

6. Cluster Analysis and Techniques

Cluster Analysis is the process of grouping objects such that objects in the same cluster are more similar to each other than to those in other clusters.

Methods:

Partitioning Methods (e.g., K-Means): Construct partitions and evaluate them by criteria.

Hierarchical Methods: Create hierarchical decomposition using top-down or bottom-up approaches.

Density-Based Methods: Identify dense regions (e.g., DBSCAN) to form clusters of arbitrary shapes.